

Predictive Maintenance for Industrial Machines

Machine Learning 1, Final Project

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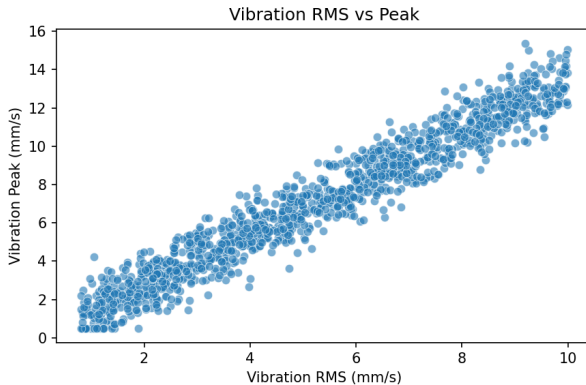
Objectives

- Provide maintenance decisions for FactoryPulse machines
- Establish a policy for FactoryPulse to react to the maintenance decision

Initial Exploratory Insights

- Machine failures only occur in a small proportion of the data (13.4%).
- Features regarding machine age, operating hours, vibration measures and prior failures are the most impactful on failure.

Vibration Analysis



The two vibration features are strongly related, which may introduce some redundancy.

Constant Predictor as a Baseline

- The baseline always predicts majority class, i.e., non-failure of the machine.
- High accuracy observed (0.84), but not representative because of class imbalance.
- Balanced accuracy of 0.5 was observed, indicating no ability to distinguish failures from non-failures.
- PR-AUC of 0.1 observed

Five base models considered

- Gaussian Naive Bayes
- Linear discriminant analysis
- K-Nearest Neighbors
- Gradient Boosting
- Logistic regression

K-Nearest Neighbors

TODO

Gradient Boosting

- Appeared to be a good fit for the problem, despite a low interpretability.
- PR-AUC of 0.3 observed, 3x the baseline performance.

Logistic Regression

TODO

A Preventive Policy

- TODO

Choice of final model

TODO

Final Evaluation Results

TODO

Limitations and what the model misses

- The model is limited by the absence of information regarding the costs of failure and maintenance of machines.
- The missed failures look like healthy machines: characteristics close to healthy machines, and with no previous failure over the last year.